

described above, this corresponds to $N_c > N_f$, $N_c = N_f$, and $N_c < N_f$, respectively.) We shall now consider each of these cases.

$$C_1 > C_2$$

Here the powers $\exp(2ia\pi T)$ with $a > 0$ have positive-definite values of R , given by Eq. (29.3.7) as $R = 2(C_1 - C_2)a$. Lorentz invariance tells us that if any term in $\mathcal{B}_\lambda$ contains a factor $W_{A\alpha L}$, then it must contain at least two of them, so the only ways to construct terms in $\mathcal{B}_\lambda$ with $R = 2$ is to have two W s and no dependence on Y or T , or one Y and no dependence on W or T , or one factor of $\exp(2i\pi T/(C_1 - C_2))$ and no dependence on W or Y :

$$\mathcal{B}_\lambda = Y f_\lambda(\Phi) + \sum_{\alpha\beta AB} \epsilon_{\alpha\beta} W_{A\alpha L} W_{B\beta B} \ell_{\lambda AB}(\Phi) + \exp\left(\frac{2i\pi T}{C_1 - C_2}\right) v_\lambda(\Phi). \quad (29.3.9)$$

Because $f_\lambda(\Phi)$ does not depend on Y or T , it can only be the tree-approximation superpotential

$$f_\lambda(\Phi) = f(\Phi), \quad (29.3.10)$$

just as in perturbation theory. Likewise, because $\ell_{\lambda AB}(\Phi)$ does not depend on Y or T it must have equal numbers of Φ s and $\Phi^\dagger$ s, so since it does not depend on $\Phi^\dagger$ it cannot depend on Φ either. Gauge invariance then requires (for a simple gauge group) that $\ell_{\lambda AB}(\Phi)$ is proportional to δ_{AB} , and since it does not depend on T or Y , the power-counting argument of Section 27.6 shows that the coefficient of δ_{AB} can only be the one-loop contribution to the running inverse-square Wilsonian gauge coupling.

To be more explicit about the running gauge coupling, recall that in non-supersymmetric gauge theories with fermions, Eq. (18.7.2) gives the one-loop renormalization group equation as

$$\lambda \frac{dg_\lambda}{d\lambda} = b g_\lambda^3, \quad (29.3.11)$$

with

$$b = -\frac{1}{4\pi^2} \left(\frac{11}{12} C_1 - \frac{1}{6} C_2 \right), \quad (29.3.12)$$

with the coefficient of C_2 taken as $-1/6$ rather than $-1/3$ because we are now counting the left-chiral states of antifermions separately from the left-chiral states of particles. As we have seen in Section 28.2, the effect of gauginos is to multiply the C_1 -term by a factor 9/11, while the effect of the scalar components of the left-chiral superfields (such as squarks and sleptons) is to multiply the C_2 -term by a factor 3/2, so in supersymmetric